DP024	**DP024**
Physics 2	Fizik 2
Semester II	Semester II
Session 2022/2023	Sesi 2022/2023
2 hours	2 jam

KOLEJ MATRIKULASI NEGERI SEMBILAN

UJIAN SELARAS 2
DP024

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Untuk Kegunaan Pemeriksa		
NO. SOALAN	**MARKAH PENUH**	**MARKAH DIPEROLEH**
1	17	
2	12	
3	9	
4	14	
5	14	
6	14	
JUMLAH	**80**	

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum *Laju cahaya dalam vakum*	c	$= 3.00 \times 10^{8}$ m s^{-1}
Permeability of free space *Ketelapan ruang bebas*	μ_o	$= 4\pi \times 10^{-7}$ H m^{-1}
Permittivity of free space *Ketelusan ruang bebas*	ε_o	$= 8.85 \times 10^{-12}$ F m^{-1}
Electron charge magnitude *Magnitud cas elektron*	e	$= 1.60 \times 10^{-19}$ C
Planck constant *Pemalar Planck*	h	$= 6.63 \times 10^{-34}$ J s
Electron mass *Jisim elektron*	m_e	$= 9.11 \times 10^{-31}$ kg $= 5.49 \times 10^{-4}$ u
Neutron mass *Jisim neutron*	m_n	$= 1.674 \times 10^{-27}$ kg $= 1.008665$ u
Proton mass *Jisim proton*	m_p	$= 1.672 \times 10^{-27}$ kg $= 1.007277$ u
Deuteron mass *Jisim deuteron*	m_d	$= 3.34 \times 10^{-27}$ kg $= 2.014102$ u
Molar gas constant *Pemalar gas molar*	R	$= 8.31$ J K^{-1} mol^{-1}
Rydberg constant *Pemalar Rydberg*	R_H	$= 1.097 \times 10^{7}$ m^{-1}
Avogadro constant *Pemalar Avogadro*	N_A	$= 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant *Pemalar Boltzmann*	k	$= 1.38 \times 10^{-23}$ J K^{-1}
Gravitational constant *Pemalar graviti*	G	$= 6.67 \times 10^{-11}$ N m^{2} kg^{-2}
Free-fall acceleration *Pecutan jatuh bebas*	g	$= 9.81$ m s^{-2}

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Atomic mass unit *Unit jisim atom*	1 u	$= 1.66 \times 10^{-27}$ kg $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt *Elektron volt*	1 eV	$= 1.6 \times 10^{-19}$ J
Constant of proportionality for Coulomb's law *Pemalar hukum Coulomb*	$k = \frac{1}{4\pi\varepsilon_o}$	$= 9.0 \times 10^{9}$ N m^{2} C^{-2}
Atmospheric pressure *Tekanan atmosfera*	1 atm	$= 1.013 \times 10^{5}$ Pa
Density of water *Ketumpatan air*	ρ_w	$= 1000$ kg m^{-3}

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

1. $x = A\sin\omega t$
2. $\omega = \frac{2\pi}{T} = 2\pi f$
3. $v = A\omega\cos\omega t$
4. $a = -A\omega^2\sin\omega t$
5. $k = \frac{2\pi}{\lambda}$
6. $v = f\lambda$
7. $y(x,t) = A\sin(\omega t \pm kx)$
8. $v = \frac{dy}{dt}$
9. $y = 2A\cos kx\sin\omega t$
10. $v = \sqrt{\frac{T}{\mu}}$
11. $I = \frac{dQ}{dt}$
12. $Q = ne$
13. $\rho = \frac{RA}{l}$
14. $R_{eff} = R_1 + R_2 + R_3 + \cdots + R_n$
15. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$
16. $V = IR$
17. $C = \frac{Q}{V}$
18. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$
19. $C_{eff} = C_1 + C_2 + C_3 + \cdots + C_n$
20. $\tau = RC$
21. $Q = Q_o e^{\frac{-t}{RC}}$
22. $Q = Q_o\left(1 - e^{\frac{-t}{RC}}\right)$
23. $B = \frac{\mu_o I}{2\pi r}$
24. $B = \frac{\mu_o NI}{L}$
25. $B = \frac{\mu_o I}{2r}$
26. $\phi = BA\cos\theta$
27. $\Phi = N\phi$
28. $\varepsilon = -\frac{d\phi}{dt}$
29. $\varepsilon = Blv\sin\theta$
30. $\varepsilon = -NA\frac{dB}{dt}$
31. $\varepsilon = -NB\frac{dA}{dt}$
32. $R = 2f$

33. $\dfrac{1}{f} = \dfrac{1}{u} + \dfrac{1}{v}$

34. $m = \dfrac{h_i}{h_o} = -\dfrac{v}{u}$

35. $y_m = \dfrac{m\lambda D}{d}$

36. $y_m = \dfrac{\left(m + \frac{1}{2}\right)\lambda D}{d}$

37. $\Delta y = \dfrac{\lambda D}{d}$

*Answer **all** questions.*

1 (a)

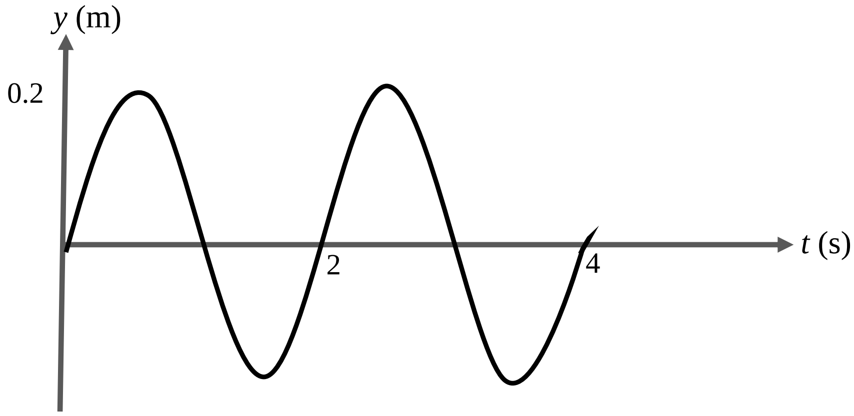

FIGURE 1

FIGURE 1 shows a graph of displacement, y versus time, t of a progressive wave propagates to the positive x-axis. If wavelength given is 0.5 m, determine the

(i) amplitude.
(ii) frequency.
(iii) wave number.
(iv) progressive wave equation.
(v) wave propagation velocity.
(vi) particle vibrational velocity at $x = 1.0$ m and $t = 2.0$ s.

[13 *marks*]

(b) Two sinusoidal waves traveling in opposite directions interfere to produce a standing wave with the wave function

$$y\,(x,t) = 1.5 \sin 0.5x \cos 220t$$

where y and x are in metre and t is in second, determine the

(i) wavelength
(ii) frequency
(iii) speed of waves

[4 *marks*]

2 (a) Current of 0.25 A flows in a lightbulb for 1 hour. Calculate the number of electrons that pass through the bulb.

[2 *marks*]

(b) (i) A thin film resistor of length 10.0 mm, width 4.0 mm and thickness 9.0 μm is made from carbon. The resistivity of carbon is 3.9×10^{-5} Ω m. Calculate the resistance.

(ii) Sketch graph potential difference, V against current, I if metallic nichrome is used instead of carbon.

[4 *marks*]

(c)

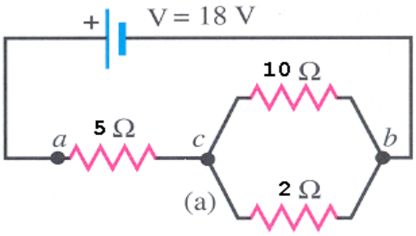

FIGURE 1

FIGURE 1 shows $V = 18$ V. Calculate for the above circuit

(i) effective resistance.
(ii) potential difference and current flow of the 10 Ω resistor.

[6 *marks*]

3 (a) Two conductors having net charges of +10.0 μC and –10.0 μC have a potential difference of 10.0 V between them. Determine the capacitance of the system.

[1 *mark*]

(b)

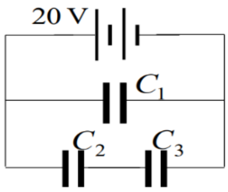

FIGURE 2

FIGURE 2 shows a combination of three capacitors where $C_1 = 100\ \mu$F, $C_2 = 22\ \mu$F and $C_3 = 47\ \mu$F. A 20 V supply is connected to the combination. Determine

(i) the effective capacitance in the circuit.
(ii) the charge stored in the capacitor C_1.

[5 *marks*]

(c) A 12 μF capacitor is charged to a potential difference of 6 V. It is then disconnected and discharges through 20 kΩ resistor. Calculate

(i) the time constant
(ii) the charge remained in the capacitor after 0.5 *s* .

[3 *marks*]

4 (a) Calculate the magnitude of magnetic field in air at a point 5 cm from a long straight wire carrying a current of 25 A.

[2 *marks*]

(b) (i) A circular coil has 30 turns and a radius of 22.5 cm. If the current flowing in the coil is 5.96 A, calculate the magnetic field strength at the centre of the coil.

(ii) A magnetic field of strength 0.5 T is directed perpendicular to the plane of a circular coil of area 2 cm^2. Calculate the magnetic flux through the area enclosed by this circular coil.

(iii) Calculate the flux linkage in a circular coil of 15 turns and area 25 cm^2 in a magnetic field of strength 5 T parallel to the plane.

[7 *marks*]

(c)

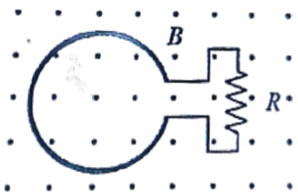

FIGURE 3

FIGURE 3 shows a circular loop connected in series with a resistor *R*, placed in a uniform magnetic field *B* with the plane of the loop perpendicular to the field. The magnetic field *B* =1.5 T, the diameter of the loop is 0.06 m and the resistance *R* = 2 Ω. If magnetic field, *B* is doubled in 0.3 s, calculate the

(i) magnitude of induced emf in the loop

(ii) Refer to c(i), calculate the current in the loop.

(iii) By using Lenz's law, state the direction of induced current in the loop. Neglect the area of the non-circular portion of the wire.

[5 *marks*]

5 (a) An object 4.0 cm high is placed 15.0 cm from a convex mirror of focal length 5.0 cm.

(i) Calculate the position of the image from the mirror.

(ii) State TWO of characteristics of image.

(iii) Calculate the magnification of the image.

[7 *marks*]

(b) A concave lens has a focal length of 8.0 cm. If the image is virtual and the image distance is one third of the object distance, calculate the

(i) object distance.

(ii) magnification of the image.

[4 *marks*]

(c) The virtual image of 50 cents coin has twice the diameter when a convex lens placed 2.82 cm from it. Calculate the focal length of the lens.

[3 *marks*]

6 (a) In a Young's double slit experiment, a yellow monochromatic light of wavelength 580 nm shines on the double slit. The separation between the slits is 0.49 mm and it is placed 1.50 m from a screen. Calculate the

(i) separation between the zeroth-order maxima and first-order maxima.

(ii) separation between the second-order maxima and fourth-order maxima.

[6 *marks*]

(b) Two slits with separation of 0.10 mm are illuminated by light 620 nm and the interference pattern is observed on screen 4.00 m from the slits. Calculate the

(i) distance of second dark fringe from central bright.

(ii) distance between the second dark fringe and third bright fringe.

(iii) fringes separation.

[8 *marks*]

END OF QUESTION PAPER